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ECONOMIC INSTRUMENTS FOR ENVIRONMENTAL REGULATION

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I. INTRODUCTION

As recently as a decade ago environmental regulators and lobbying groups with a special interest in environmental protection looked upon the market system as a powerful adversary. That the market unleashed powerful forces was widely recognized and that those forces clearly acted to degrade the environment was widely lamented. Conflict and confrontation became the battle cry for those groups seeking to protect the environment as they set out to block market forces whenever possible.

Among the more enlightened participants in the environmental policy process the air of confrontation and conflict has now begun to recede in many parts of the world. Leading environmental groups and regulators have come to realize that the power of the market can be harnessed and channelled toward the achievement of environmental goals, through an economic incentives approach to regulation. Forward-looking business people have come to appreciate the fact that cost-effective regulation can make them more competitive in the global market-place than regulations which impose higher-than-necessary control costs.

The change in attitude has been triggered by a recognition that this former adversary, the market, can be turned into a powerful ally. In contrast to the traditional regulatory approach, which makes mandatory particular forms of behaviour or specific technological choices, the economic incentive approach allows more flexibility in how the environmental goal is reached. By changing the incentives an individual agent faces, the best private choice can be made to coincide with the best social choice. Rather than relying on the regulatory authority to identify the best course of action, the individual agent can use his or her typically superior information to select the best means of meeting an assigned emission reduction responsibility. This flexibility achieves environmental goals at lower cost, which, in turn, makes the goals easier to achieve and easier to establish.

One indicator of the growing support for the use of economic incentive approaches for environmental control in the United States is the favourable treatment it has recently received both in the popular business[1] and environmental[2] press. Some public interest environmental organizations have now even adopted economic incentive approaches as a core part of their strategy for protecting the environment.[3]

In response to this support the emissions trading concept has recently been applied to reducing the lead content in gasoline, to controlling both ozone depletion and non-point sources of water pollution, and was also prominently featured in the Bush administration proposals for reducing acid rain and smog unveiled in June 1989.

Our knowledge about economic incentive approaches has grown rapidly in the two decades in which they have received serious analytical attention. Not only have the theoretical models become more focused and the empirical work more detailed, but we have now had over a decade of experience with emissions trading in the US and emission charges in Europe.

As the world community becomes increasingly conscious of both the need to tighten environmental controls and the local economic perils associated with tighter controls in a highly competitive global market-place, it seems a propitious time to stand back and to organize what we have learned about this practical and promising approach to pollution control that may be especially relevant to current circumstances. In this paper I will draw upon economic theory, empirical studies, and actual experience with implementation to provide a brief overview of some of the major lessons we have learned about two economic incentive approaches—emissions trading and emission charges—as well as their relationships to the more traditional regulatory policy.[4]

II. THE POLICY CONTEXT

(i) Emissions Trading

Stripped to its bare essentials, the US Clean Air Act[5] relies upon a *command-and-control* approach to controlling pollution. Ambient standards establish the highest allowable concentration of the pollutant in the ambient air for each conventional pollutant. To reach these prescribed ambient standards, emission standards (legal emission ceilings) are imposed on a large number of specific emission points such as stacks, vents, or storage tanks. Following a survey of the technological options of control, the control authority selects a favoured control technology and calculates the amount of emission reduction achievable by that technology as the basis for setting the emission standard. Technologies yielding larger amounts of control (and, hence, supporting more stringent emission standards) are selected for new emitters and for existing emitters in areas where is it very difficult to meet the ambient standard. The responsibility for defining and enforcing these standards is shared in legislatively specified ways between the national government and the various state governments.

The emissions trading programme attempts to inject more flexibility into the manner in which the objectives of the Clean Air Act are met by allowing sources a much wider range of choice in how they satisfy their legal pollution control responsibilities than possible in the command-and-control approach. Any source choosing to reduce emissions at any discharge point more than required by its emission standard can apply to the control authority for certification of the excess control as an 'emission reduction credit' (ERC). Defined in terms of a specific amount of a particular pollutant, the certified emissions reduction credit can be used to satisfy emission standards at other (presumably more expensive to control) discharge points controlled by the creating source or it can be sold to

[1] See, for example, Main (1988).
[2] See, for example, Stavins (1989).
[3] See the various issues in Volume XX of the EDF Letter, a report to members of the Environmental Defense Fund.
[4] In the limited space permitted by this paper only a few highlights can be illustrated. All of the details of the proofs and the empirical work can be found in the references listed at the end of the paper. For a comprehensive summary of this work see Tietenberg (1980), Liroff (1980), Bohm and Russell (1985), Tietenberg (1985), Liroff (1986), Dudek and Palmisano (1988), Hahn (1989), Hahn and Hester (1989*a* and 1989*b*), and Tietenberg (1989*b*).
[5] The US Clean Air Act (42 U.S.C. 7401–642) was first passed in 1955. The central thrust of the approach described in this paragraph was initiated by the Clean Air Act Amendments of 1970 with mid-course corrections provided by the Clean Air Act Amendments of 1977.

other sources. By making these credits transferable, the US Environmental Protection Agency (EPA) has allowed sources to find the cheapest means of satisfying their requirements, even if the cheapest means are under the control of another firm. The ERC is the currency used in emissions trading, while the offset, bubble, emissions banking, and netting policies govern how this currency can be stored and spent.[6]

The *offset policy* requires major new or expanding sources in 'non-attainment' areas (those areas with air quality worse than the ambient standards) to secure sufficient offsetting emission reductions (by acquiring ERCs) from existing firms so that the air is cleaner after their entry or expansion than before.[7] Prior to this policy no new firms were allowed to enter non-attainment areas on the grounds they would interfere with attaining the ambient standards. By introducing the offset policy EPA allowed economic growth to continue while assuring progress toward attainment.

The *bubble policy* receives its unusual name from the fact that it treats multiple emission points controlled by existing emitters (as opposed to those expanding or entering an area for the first time) as if they were enclosed in a bubble. Under this policy only the total emissions of each pollutant leaving the bubble are regulated. While the total leaving the bubble must be not larger than the total permitted by adding up all the corresponding emission standards within the bubble (and in some cases the total must be 20 per cent lower), emitters are free to control some discharge points less than dictated by the corresponding emission standard as long as sufficient compensating ERCs are obtained from other discharge points within the bubble. In essence sources are free to choose the mix of control among the discharge points as long as the overall emission reduction requirements are satisfied. Multi-plant bubbles are allowed, opening the possibility for trading ERCs among very different kinds of emitters.

Netting allows modifying or expanding sources (but not new sources) to escape from the need to meet the requirements of the rather stringent new source review process (including the need to acquire offsets) so long as any net increase in emissions (counting any ERCs earned elsewhere in the plant) is below an established threshold. In so far as it allows firms to escape particular regulatory requirements by using ERCs to remain under the threshold which triggers applicability, netting is more properly considered regulatory relief than regulatory reform.

Emissions banking allows firms to store certified ERCs for subsequent use in the offset, bubble, or netting programmes or for sale to others.

Although comprehensive data on the effects of the programme do not exist because substantial proportions of it are administered by local areas and no one collects information in a systematic way, some of the major aspects of the experience are clear.[8]

• The programme has unquestionably and substantially reduced the costs of complying with the requirements of the Clean Air Act. Most estimates place the accumulated capital savings for all components of the programme at over $10 billion. This does not include the recurring savings in operating cost. On the other hand the programme has not produced the magnitude of cost savings that was anticipated by its strongest proponents at its inception.

• The level of compliance with the basic provisions of the Clean Air Act has increased. The emissions trading programme increased the possible means for compliance and sources have responded.

• Somewhere between 7,000 and 12,000 trading transactions have been consummated. Each of these transactions was voluntary and for the participants represented an improvement over the traditional regulatory approach. Several of these transactions involved the introduction of innovative control technologies.

• The vast majority of emissions trading transactions have involved large pollution sources trading emissions reduction credits either created by excess

[6] The details of this policy can be found in 'Emissions Trading Policy Statement' 51 *Federal Register* 43829 (4 December 1986).

[7] Offsets are also required for major modifications in areas which have attained the standards if the modifications jeopardize attainment.

[8] See, for example, Tietenberg (1985), Hahn and Hester (1989*a* and 1989*b*), and Dudek and Palmisano (1988).

control of uniformly mixed pollutants (those for which the location of emission is not an important policy concern) or involving facilities in close proximity to one another.

• Though air quality has certainly improved for most of the covered pollutants, it is virtually impossible to say how much of the improvement can be attributed to the emissions trading programme. The emissions trading programme complements the traditional regulatory approach, rather than replaces it. Therefore, while it can claim to have hastened compliance with the basic provisions of the act and in some cases to have encouraged improvements beyond the act, improved air quality resulted from the package taken together, rather than from any specific component.

(ii) Emissions Charges

Emission charges are used in both Europe and Japan, though more commonly to control water pollution than air pollution.[9] Currently effluent charges are being used to control water pollution in France, Italy, Germany, and the Netherlands. In both France and the Netherlands the charges are designed to raise revenue for the purpose of funding activities specifically designed to improve water quality.

In Germany dischargers are required to meet minimum standards of waste water treatment for a number of defined pollutants. Simultaneously a fee is levied on every unit of discharge depending on the quantity and noxiousness of the effluent. Dischargers meeting or exceeding state-of-the-art effluent standards have to pay only half the normal rate.

The Italian effluent charge system was mainly designed to encourage polluters to achieve provisional effluent standards as soon as possible. The charge is nine times higher for firms that do not meet the prescribed standards than for firms that do meet them. This charge system was designed only to facilitate the transition to the prescribed standards so it is scheduled to expire once full compliance has been achieved.[10]

Air pollution emission charges have been implemented by France and Japan. The French air pollution charge was designed to encourage the early adoption of pollution control equipment with the revenues returned to those paying the charge as a subsidy for installing the equipment. In Japan the emission charge is designed to raise revenue to compensate victims of air pollution. The charge rate is determined primarily by the cost of the compensation programme in the previous year and the amount of remaining emissions over which this cost can be applied *pro rata*.

Charges have also been used in Sweden to increase the rate at which consumers would purchase cars equipped with a catalytic converter. Cars not equipped with a catalytic converter were taxed, while new cars equipped with a catalytic converter were subsidized.

While data are limited a few highlights seem clear:

• Economists typically envisage two types of effluent or emissions charges. The first, an efficiency charge, is designed to produce an efficient outcome by forcing the polluter to compensate completely for all damage caused. The second, a cost-effective charge, is designed to achieve a predefined ambient standard at the lowest possible control cost. In practice, few, if any, implemented programmes fit either of these designs.

• Despite being designed mainly to raise revenue, effluent charges have typically improved water quality. Though the improvements in most cases have been small, apparently due to the low level at which the effluent charge rate is set, the Netherlands, with its higher effective rates, reports rather large improvements. Air pollution charges typically have not had much effect on air quality because the rates are too low and, in the case of France, most of the revenue is returned to the polluting sources.

[9] See Anderson (1977), Brown and Johnson (1984), Bressers (1988), Vos (1989), Opschoor and Vos (1989), and Sprenger (1989).

[10] The initial deadline for expiration was 1986, but it has since been postponed.

• The revenue from charges is typically earmarked for specific environmental purposes rather than contributed to the general revenue as a means of reducing the reliance on taxes that produce more distortions in resource allocation.

• The Swedish tax on heavily polluting vehicles and subsidy for new low polluting vehicles was very successful in introducing low polluting vehicles into the automobile population at a much faster than normal rate. The policy was not revenue neutral, however; owing to the success of the programme in altering vehicle choices, the subsidy payments greatly exceeded the tax revenue.

III. FIRST PRINCIPLES

Theory can help us understand the characteristics of these economic approaches in the most favourable circumstances for their use and assist in the process of designing the instruments for maximum effectiveness. Because of the dualistic nature of emission charges and emission reduction credits,[11] implications about emission charges and emissions trading flow from the same body of theory.

Drawing conclusions about either of these approaches from this type of analysis, however, must be done with care because operational versions typically differ considerably from the idealized versions modelled by the theory. For example, not all trades that would be allowed in an ideal emissions trading programme are allowed in the current US emissions trading programme. Similarly the types of emissions charges actually imposed differ considerably from their ideal versions, particularly in the design of the rate structure and the process for adjusting rates over time.

Assuming all participants are cost-minimizers, a 'well-defined' emissions trading or emission charge system could cost-effectively allocate the control responsibility for meeting a predefined pollution target among the various pollution sources despite incomplete information on the control possibilities by the regulatory authorities.[12]

The intuition behind this powerful proposition is not difficult to grasp. Cost-minimizing firms seek to minimize the sum of (*a*) either ERC acquisition costs or payments of emission charges and (*b*) control costs. Minimization will occur when the marginal cost of control is set equal to the emission reduction credit price or the emission charge. Since all cost-minimizing sources would choose to control until their marginal control costs were equal to the same price or charge, marginal control costs would be equalized across all discharge points, precisely the condition required for cost-effectiveness.[13]

Emission charges could also sustain a cost effective allocation of the control responsibility for meeting a predefined pollution target, but only if the control authority knew the correct level of the charge to impose or was willing to engage in an iterative trial-and-error process over time to find the correct level. Emissions trading does not face this problem because the price level is established by the market, not the control authority.[14]

Though derived in the rarified world of theory, the practical importance of this theorem should not be underestimated. Economic incentive approaches offer a unique opportunity for regulators to solve a fundamental dilemma. The control authorities' desire to allocate the responsibility for control cost-effectively is inevitably frustrated by a lack of information sufficient to achieve this objective. Economic incentive approaches create a system of incentives in which those who have the best knowledge about control opportunities, the environmental managers for the industries, are encouraged to use

[11] Under fairly general conditions any allocation of control responsibility achieved by an emissions trading programme could also be achieved by a suitably designed system of emission charges and vice versa.

[12] For the formal demonstration of this proposition see Baumol and Oates (1975), Montgomery (1972), and Tietenberg (1985).

[13] It should be noted that while the allocation is cost-effective, it is not necessarily efficient (the amount of pollution indicated by a benefit–cost comparison). It would only be efficient if the predetermined target happened to coincide with the efficient amount of pollution. Nothing guarantees this outcome.

[14] See Tietenberg (1988) for a more detailed explanation of this point.

that knowledge to achieve environmental objectives at minimum cost. Information barriers do not preclude effective regulation.

What constitutes a 'well-defined' emissions trading or emission charge system depends crucially on the attributes of the pollutant being controlled.[15]

To be consistent with a cost-effective allocation of the control responsibility, the policy instruments would have to be defined in different ways for different types of pollutants. Two differentiating characteristics are of particular relevance. Approaches designed to control pollutants which are uniformly mixed in the atmosphere (such as volatile organic compounds, one type of precursor for ozone formation) can be defined simply in terms of a rate of emissions flow per unit time. Economic incentive approaches sharing this design characteristic are called *emission trades* or *emission charges*.

Instrument design is somewhat more difficult when the pollution target being pursued is defined in terms of concentrations measured at a number of specific receptor locations (such as particulates). In this case the cost-effective trade or charge design must take into account the *location* of the emissions (including injection height) as well as the *magnitude* of emissions. As long as the control authorities can define for each emitter a vector of transfer coefficients, which translate the effect of a unit increase of emissions by that emitter into an increase in concentration at each of the affected receptors, receptor-specific trades or charges can be defined which will allocate the responsibility cost-effectively. The design which is consistent with cost-effectiveness in this context is called an *ambient trade* or an *ambient charge*.

Unfortunately, while the design of the ambient ERC is not very complicated,[16] implementing the markets within which these ERCs would be traded is rather complicated. In particular for each unit of planned emissions an emitter would have to acquire separate ERCs for each affected receptor. When the number of receptors is large, the result is a rather complicated set of transactions. Similarly, establishing the correct rate structure for the charges in this context is particularly difficult because the set of charges which will satisfy the ambient air quality constraints is not unique; even a trial-and-error system would not necessarily result in the correct matrix of ambient charges being put into effect.

As long as markets are competitive and transactions costs are low, the trading benchmark in an emissions trading approach does not affect the ultimate cost-effective allocation of control responsibility. When markets are non-competitive or transactions costs are high, however, the final allocation of control responsibility is affected.[17] Emission charge approaches do not face this problem.

Once the control authority has decided how much pollution of each type will be allowed, it must then decide how to allocate the operating permits among the sources. In theory emission reduction credits could either be auctioned off, with the sources purchasing them from the control authority at the market-clearing price, or (as in the US programme) created by the sources as surplus reductions over and above a predetermined set of emission standards. (Because this latter approach favours older sources over newer sources, it is known as 'grandfathering'.) The proposition suggests that either approach will ultimately result in a cost-effective allocation of the control responsibility among the various polluters as long as they are all price-takers, transactions costs are low, and ERCs are fully transferable. Any allocation of emission standards in a grandfathered approach is compatible with cost-effectiveness because the after-market in which firms can buy or sell ERCs corrects any problems with the initial allocation. This is a significant finding because it implies that under the right conditions the control authority can use this initial allocation of emissions standards to pursue distributional goals without interfering with cost-effectiveness.

When firms are price-setters rather than price-takers, however, cost-effectiveness will only be achieved if the control authority initially allocates the emission standards so a cost-effective allocation would be achieved even in the absence of any trading. (Implementing this particular allocation would, of course, require regulators to have com-

[15] For the technical details supporting this proposition see Montgomery (1972), and Tietenberg (1985).

[16] Each permit allows the holder to degrade the concentration level at the corresponding receptor by one unit.

[17] See Hahn (1984) for the mathematical treatment of this point. Further discussions can be found in Tietenberg (1985) and Misiolek and Elder (1989).

plete information on control costs for all sources, an unlikely prospect.) In this special case cost-effectiveness would be achieved even in the presence of one or more price-setting firms because no trading would take place, eliminating the possibility of exploiting any market power.

For all other emission standard assignments an active market would exist, offering the opportunity for price-setting behaviour. The larger is the deviation of the price setting source's emission standard from its cost-effective allocation, the larger is the deviation of ultimate control costs from the least-cost allocation. When the price-setting source is initially allocated an insufficiently stringent emission standard, it can inflict higher control costs on others by withholding some ERCs from the market. When an excessively stringent emission standard is imposed on a price-setting source, however, it necessarily bears a higher control cost as the means of reducing demand (and, hence, prices) for the ERCs.

Similar problems exist when transactions costs are high. High transactions costs preclude or reduce trading activity by diminishing the gains from trade. When the costs of consummating a transaction exceed its potential gains, the incentive to participate in emissions trading is lost.

IV. LESSONS FROM EMPIRICAL RESEARCH

A vast majority, though not all, of the relevant empirical studies have found the control costs to be substantially higher with the regulatory command-and-control system than the least cost means of allocating the control responsibility.

While theory tells us unambiguously that the command-and-control system will not be cost-effective except by coincidence, it cannot tell us the magnitude of the excess costs. The empirical work cited in Table 1 adds the important information that the excess costs are typically very large.[18] This is an important finding because it provides the motivation for introducing a reform programme; the potential social gains (in terms of reduced control cost) from breaking away from the status quo are sufficient to justify the trouble. Although the estimates of the excess costs attributable to a command and control presented in Table 1 overstate the cost savings that would be achieved by even an ideal economic incentive approach (a point discussed in more detail below), the general conclusion that the potential cost savings from adopting economic incentive approaches are large seems accurate even after correcting for overstatement.

Economic incentive approaches which raise revenue (charges or auction ERC markets) offer an additional benefit—they allow the revenue raised from these policies to substitute for revenue raised in more traditional ways. Whereas it is well known that traditional revenue-raising approaches distort resource allocation, producing inefficiency, economic incentive approaches enhance efficiency. Some empirical work based on the US economy suggests that substituting economic incentive means of raising revenue for more traditional means could produce significant efficiency gains.[19]

When high degrees of control are necessary, ERC prices or charge levels would be correspondingly high. The financial outlays associated with acquiring ERCs in an auction market or paying charges on uncontrolled emissions would be sufficiently large that sources would typically have lower financial burdens with the traditional command-and-control approach than with these particular economic incentive approaches. Only a 'grandfathered' trading system would guarantee that sources would be no worse off than under the command-and-control system.[20]

Financial burden is a significant concern in a highly competitive global market-place. Firms bearing

[18] A value of 1.0 in the last column of Table 1 would indicate that the traditional regulatory approach was cost-effective. A value of 4.0 would indicate that the traditional regulatory approach results in an allocation of the control responsibility which is four times as expensive as necessary to reach the stipulated pollution target.

[19] See Terkla (1984).

[20] See Atkinson and Tietenberg (1982, 1984), Hahn (1984), Harrison (1983), Krupnick (1986), Lyon (1982), Palmer *et al.*(1980), Roach *et al.* (1981), Seskin *et al.* (1983), and Shapiro and Warhit (1983) for the individual studies, and Tietenberg (1985) for a summary of the evidence.

Table 1
Empirical Studies of Air Pollution Control

Study	Pollutants Covered	Geographic Area	CAC Benchmark	Ratio of CAC Cost to Least Cost
Atkinson and Lewis	Particulates	St Louis	SIP regulations	6.00[a]
Roach *et al.*	Sulphur dioxide	Four corners in Utah	SIP regulations Colorado, Arizona, and New Mexico	4.25
Hahn and Noll	Sulphates standards	Los Angeles	California emission	1.07
Krupnick	Nitrogen dioxide regulations	Baltimore	Proposed RACT	5.96[b]
Seskin *et al.*	Nitrogen dioxide regulations	Chicago	Proposed RACT	14.40[b]
McGartland	Particulates	Baltimore	SIP regulations	4.18
Spofford	Sulphur Dioxide	Lower Delaware Valley	Uniform percentage regulations	1.78
	Particulates	Lower Delaware Valley	Uniform percentage regulations	22.00
Harrison	Airport noise	United States	Mandatory retrofit	1.72[c]
Maloney and Yandle	Hydrocarbons	All domestic DuPont plants	Uniform percentage reduction	4.15[d]
Palmer *et al.*	CFC emissions from non-aerosol applications	United States	Proposed emission standards	1.96

Notes:

CAC = command and control, the traditional regulatory approach.

SIP = state implementation plan.

RACT = reasonably available control technologies, a set of standards imposed on existing sources in non-attainment areas.

[a] Based on a 40 $\mu g/m^3$ at worst receptor.

[b] Based on a short-term, one-hour average of 250 $\mu g/m^3$.

[c] Because it is a benefit–cost study instead of a cost-effectiveness study, the Harrison comparison of the command-and-control approach with the least-cost allocation involves different benefit levels. Specifically, the benefit levels associated with the least-cost allocation are only 82 per cent of those associated with the command-and-control allocation. To produce cost estimates based on more comparable benefits, as a first approximation the least-cost allocation was divided by 0.82 and the resulting number was compared with the command-and-control cost.

[d] Based on 85 per cent reduction of emissions from all sources.

large financial burdens would be placed at a competitive disadvantage when forced to compete with firms not bearing those burdens. Their costs would be higher.

From the point of view of the source required to control its emissions, two components of financial burden are significant: (*a*) control costs and (*b*) expenditures on permits or emission charges. While only the former represent real resource costs to society as a whole (the latter are merely transferred from one group in society to another), both represent a financial burden to the source. The empirical evidence suggests that when an auction market is used to distribute ERCs (or, equivalently, when all uncontrolled emissions are subject to an emissions charge), the ERC expenditures (charge outlays) would frequently be larger in magnitude than the control costs; the sources would spend more on ERCs (or pay more in charges) than they would on the control equipment. Under the traditional command-and-control system firms make no financial outlays to the government. Although control costs are necessarily higher with the command-and-control system than with an economic incentive approach, they are not so high as to outweigh the additional financial outlays required in an auction market permit system (or an emissions tax system). For this reason existing sources could be expected vehemently to oppose an auction market or emission charges despite their social appeal, unless the revenue derived is used in a manner which is approved by the sources, and the sources with which it competes are required to absorb similar expenses. When environmental policies are not co-ordinated across national boundaries, this latter condition would be particularly difficult to meet.

In the absence of either a politically popular way to use the revenue or assurances that competitors will face similar financial burdens, this political opposition could be substantially reduced by grandfathering. Under grandfathering, sources have only to purchase any additional ERCs they may need to meet their assigned emission standard (as opposed to purchasing sufficient ERCs or paying charges to cover all uncontrolled emissions in an auction market). Grandfathering is *de facto* the approach taken in the US emissions trading programme.

Grandfathering has its disadvantages. Because ERCs become very valuable, especially in the face of stringent air quality regulations, sources selling emission reduction credits would be able to command very high prices. By placing heavy restrictions on the amount of emissions, the control authority is creating wealth for existing firms *vis-à-vis* new firms.

Although reserving some ERCs for new firms is possible (by assigning more stringent emission standards than needed to reach attainment and using the 'surplus' air quality to create government-held ERCs), this option is rarely exercised in practice. In the United States under the offset policy firms typically have to purchase sufficient ERCs to more than cover all uncontrolled emissions, while existing firms only have to purchase enough to comply with their assigned emission standard. Thus grandfathering imposes a bias against new sources in the sense that their financial burden is greater than that of an otherwise identical existing source, even if the two sources install exactly the same emission control devices. This new source bias could retard the introduction of new facilities and new technologies by reducing the cost advantage of building new facilities which embody the latest innovations.

While it is clear from theory that larger trading areas offer the opportunities for larger potential cost savings in an emissions trading programme, some empirical work suggests that substantial savings can be achieved in emissions trading even when the trading areas are rather small.

The point of this finding is *not* that small trading areas are fine; they do retard progress toward the standard. Rather, when political considerations allow only small trading areas or nothing, emissions trading still can play a significant role.

Sometimes political considerations demand a trading area which is smaller than the ideal design. Whether large trading areas are essential for the effective use of this policy is therefore of some relevance. In general, the larger the trading area, the larger would be the potential cost savings due to a wider set of cost reduction opportunities that would become available. The empirical question is how sensitive the cost estimates are to the size of the trading areas.

One study of utilities found that even allowing a plant to trade among discharge points within that plant could save from 30 to 60 per cent of the costs of complying with new sulphur oxide reduction regulations, compared to a situation where no trading whatsoever was permitted.[21] Expanding the trading possibilities to other utilities within the same state permitted a further reduction of 20 per cent, while allowing interstate trading permitted another 15 per cent reduction in costs. If this study is replicated in other circumstances, it would appear that even small trading areas offer the opportunity for significant cost reduction.[22]

Although only a few studies of the empirical impact of market power on emissions trading have been accomplished, their results are consistent with a finding that market power does not seem to have a large effect on regional control costs in most realistic situations.[23]

Even in areas having especially stringent controls, the available evidence suggests that price manipulation is not a serious problem. In an auction market the price-setting source reduces its financial burden by purchasing fewer ERCs in order to drive the price down. To compensate for the smaller number of ERCs purchased, the price-setting source must spend more on controlling its own pollution, limiting the gains from price manipulation. Although these actions could have a rather large impact on *regional financial burden*, they would under normal circumstances have a rather small effect on *regional control costs*. Estimates typically suggest that control costs would rise by less than 1 per cent if market power were exercised by one or more firms.

It should not be surprising that price manipulation could have rather dramatic effects on regional financial burden in an auction market, since the cost of *all* ERCs is affected, not merely those purchased by the price-setting source. The perhaps more surprising result is that control costs are quite insensitive to price-setting behaviour. This is due to the fact that the only control cost change is the net difference between the new larger control burden borne by the price searcher and the correspondingly smaller burden borne by the sources having larger-than-normal allocations of permits. Only the costs of the marginal units are affected.

Within the class of grandfathered distribution rules, some emission standard allocations create a larger potential for strategic price behaviour than others. In general the larger the divergence between the control responsibility assigned to the price-searching source by the emission standards and the cost-effective allocation of control responsibility, the larger the potential for market power. When allocated too little responsibility by the control authority, price-searching firms can exercise power on the selling side of the market, and when allocated too much, they can exercise power on the buying side of the market.

According to the existing studies it takes a rather considerable divergence from the cost-effective allocation of control responsibility to produce much difference in regional control costs. In practice the deviations from the least cost allocation caused by market power pale in comparison to the much larger potential cost reductions achievable by implementing emissions trading.[24]

V. LESSONS FROM IMPLEMENTATION

Though the number of transactions consummated under the Emissions Trading Program has been large, it has been smaller than expected. Part of this failure to fulfil expectations can be explained as the result of unrealistically inflated expectations. More restrictive regulatory decisions than expected and

[21] ICF, Inc. (1989).

[22] As indicated below, the fact that so many emissions trades have actually taken place within the same plant or among contiguous plants provides some confirmation for this result.

[23] For individual studies see de Lucia (1974), Hahn (1984), Stahl, Bergman and Mäler (1988), and Maloney and Yandle (1984). For a survey of the evidence see Tietenberg (1985).

[24] Strategic price behaviour is not the only potential source of market power problems. Firms could conceivably use permit markets to drive competitors out of business. See Misiolek and Elder (1989). For an analysis which concludes that this problem is relatively rare and can be dealt with on a case-by-case basis should it arise, see Tietenberg (1985).

higher than expected transaction costs also bear some responsibility.

The models used to calculate the potential cost savings were not (and are not) completely adequate guides to reality. The cost functions in these models are invariably *ex ante* cost functions. They implicitly assume that the modelled plant can be built from scratch and can incorporate the best technology. In practice, of course, many existing sources cannot retrofit these technologies and therefore their *ex post* control options are much more limited than implied by the models.

The models also assume all trades are multilateral and are simultaneously consummated, whereas actual trades are usually bilateral and sequential. The distinction is important for non-uniformly mixed pollutants;[25] bilateral trades frequently are constrained by regulatory concerns about decreasing air quality at the site of the acquiring source. Because multilateral trades would typically incorporate compensating reductions coming from other nearby sources, these concerns normally do not arise when trades are multilateral and simultaneous. In essence the models implicitly assume an idealized market process, which is only remotely approximated by actual transactions.

In addition some non-negligible proportion of the expected cost savings recorded by the models for non-uniformly mixed pollutants is attributable to the substantially larger amounts of emissions allowed by the modelled permit equilibrium.[26] For example, the cost estimates imply that the control authority is allowed to arrange the control responsibility in *any* fashion that satisfies the ambient air quality standards. In practice the models allocate more uncontrolled emissions to sources with tall stacks because those emissions can be exported. Exported emissions avoid control costs without affecting the readings at the local monitors. That portion of the cost savings estimated by the models in Table 1 which is due to allowing increased emissions is not acceptable to regulators. Some recent work has suggested that the benefits received from the additional emission control required by the command and control approach may be justified by the net benefits received.[27] The regulatory refusal to allow emission increases was apparently consistent with efficiency,[28] but it was not consistent with the magnitiude of cost savings anticipated by the models.

Certain types of trades assumed permissible by the models are prohibited by actual trading rules. New sources, for example, are not allowed to satisfy the New Source Performance Standards (which imply a particular control technology) by choosing some less stringent control option and making up the difference with acquired emission reduction credits; they must install the degree of technological control necessary to meet the standard. Typically this is the same technology used by EPA to define the standard in the first place.

A lot of uncertainty is associated with emission reduction credit transactions since they depend so heavily on administrative action. All trades must be approved by the control authorities. If the authorities are not co-operative or at least consistent, the value of the created emission reduction credits could be diminished or even destroyed.

For non-uniformly mixed pollutants, trades between geographically separated sources will only be approved after dispersion modelling has been accomplished by the applicants. Not only is this modelling expensive, it frequently ends up raising questions which ultimately lead to the transaction being denied. Few trades requiring this modelling have been consummated.

Trading activity has also been inhibited by the paucity of emission banks. The US system allows states to establish emission banks, but does not require them to do so. As of 1986 only seven of the fifty states had established these banks. For sources in the rest of the states the act of creating emission credits is undervalued because the credits cannot be

[25] See Tietenberg and Atkinson (1989) for a demonstration that this is an empirically significant point.

[26] This is demonstrated in Atkinson and Tietenberg (1987).

[27] See Oates, Portney, and McGartland (1988).

[28] Not all of the cost savings, of course, is due to the capability to increase emissions. The remaining portion of the savings, which is due to taking advantage of opportunities to control a given level of emissions at a lower cost, is still substantial and can be captured by a well-designed permit system which does not allow emissions to increase beyond the command-and-control benchmark. See the calculations in Atkinson and Tietenberg (1987).

legally held for future use. The supply of emission reduction credits is hence less than would be estimated by the models.

The Emissions Trading Program seems to have worked particularly well for trades involving uniformly mixed pollutants and for trades of non-uniformly mixed pollutants involving contiguous discharge points.

It is not surprising that most consummated trades have been internal (where the buyer and seller share a common corporate parent) rather than external. Not only are the uncertainties associated with inter-firm transfers avoided, but most internal trades involve contiguous facilities. Trades between contiguous facilities do not trigger a requirement for dispersion modelling.[29]

It is also not surprising that the plurality of consummated trades involve volatile organic compounds, which are uniformly mixed pollutants. Since dispersion modelling is not required for uniformly mixed pollutants even when the trading sources are somewhat distant from one another, trades involving these pollutants are cheaper to consummate. Additionally emissions trades involving uniformly mixed pollutants do not jeopardize local air quality since the location of the emissions is not a matter of policy consequence.

The establishment of the Emissions Trading Program has encouraged technological progress in pollution control. Although generally the degree of progress has been modest, it has been more dramatic in areas where emission reductions have been sufficiently stringent as to restrict the availability of emission reduction credits created by more traditional means.[30]

Theory would lead us to expect more technological progress with emissions trading than with a command-and-control policy because it changes the incentives so drastically. Under a command-and-control approach technological changes discovered by the control authority typically lead to more stringent standards (and higher costs) for the sources. Sources have little incentive to innovate and a good deal of incentive to hide potential innovations from the control authority. With emissions trading, on the other hand, innovations allowing excess reductions create saleable emission reduction credits.

The evidence suggests that the expectations based on this theory have been borne out to a limited degree in the operating programme. The most prominent example of technological change has been the substitution of water-based solvents for solvents containing volatile organic compounds. Though somewhat more expensive, this substitution made economic sense once the programme was introduced.

It should probably not be surprising that the number of new innovations stimulated by the programme is rather small. As long as cheaper ways of creating credits within existing processes (fuel substitution, for example) are available, it would be unreasonable to expect large investments in new technologies with unproven reliabilities. On the other hand as the degree of control rises and the supply of readily available credits dries up, the demand for new technologies would be expected to rise as well. This expectation seems to have been borne out in those areas where unusually low air quality or stringent regulatory rules have served to limit the available credits.[31]

This is an important point. Those who fail to consider the dynamic advantages of an economic incentive approach sometimes suggest that if few credits would be traded, implementing a system of this type has no purpose. In fact it has a substantial purpose—the encouragement of new technologies to meet the increasingly stringent standards.

Introducing the Emissions Trading Program has provided an opportunity to control sources which can reduce emissions relatively cheaply, but which under the traditional policy were under-regulated due either to their financially precarious position or the fact that they were not subject to regulation.[32]

[29] The fact that so many trades have taken place between contiguous discharge points serves as confirmation that substantial savings can be achieved even if the geographic boundaries of the trading area are quite restricted.

[30] For more details see Tietenberg (1985), Maleug (1989), and Dudek and Palmisano (1988).

[31] For the experience in California see Dudek and Palmisano (1988).

[32] See Tietenberg (1985).

Due to the social distress caused by any resulting unemployment, the control authorities and the courts are understandably reluctant to enforce stringent emission standards against firms which would not be able to pass higher costs on to customers without considerable loss of production. Since many of these sources could control emissions at a lower marginal cost than other sources, their political immunity from control makes regional control costs higher than necessary; other sources have to control their own emissions to a higher degree (at a higher marginal cost) to compensate.

Due to its ability to separate the issue of who pays for the reduction from the issue of which discharge points are to be controlled, the emissions trading programme provides a way to secure those low cost reductions. The command-and-control policy would assign, as normal, a very low (perhaps zero) emission reduction to any previously unregulated firm. Once emissions trading had been established, however, it would be in the interest of this firm to control emissions further, selling the resulting emission reduction credits. As long as the revenues from the sale at least covered the cost, this transaction could profit, or at least not hurt, the seller. Because these reductions could be achieved at a lower cost than ratcheting up the degree of control on already heavily controlled sources, non-immune sources would find purchasing the credits cheaper than controlling their own emissions to a higher degree. Everyone benefits from controlling these previously under-regulated sources.

Another unique attribute of an emissions trading approach is the capability it offers sources for leasing credits.[33]

Leasing offers an enormously useful degree of flexibility which is not available with other policy approaches to pollution control. The usefulness of leasing derives from the fact that some sources, utilities in particular, have patterns of emission that vary over time while allowable emissions remain constant. In a typical situation, for example, suppose an older utility would, in the absence of control, be emitting heavily. In the normal course of a utility expansion cycle the older plant would subsequently experience substantially reduced emissions when the utility constructed a new plant and shifted a major part of the load away from the older plant to the new plant. Ultimately growth in demand on the system would increase the emissions again for the older plant as its capacity would once again be needed. The implication of this temporal pattern is that during the middle period, as its own emissions fell well below allowable emissions, this utility could lease excess emission credits to another facility, recalling them as its own need rose with demand growth. Indeed one empirical study of the pattern of the utility demand for and supply of acid rain reduction credits over time suggests that leasing is a critical component of any cost-effective control strategy, a component that neither the traditional approach nor emission charges can offer.[34]

Leasing also provides a way for about-to-be-retired sources to participate in the reduction programme. Under the traditional approach once the deadline for compliance had been reached the utility would either have to retire the unit early or to install expensive control equipment which would be rendered useless once the unit was retired. By leasing credits for the short period to retirement, the unit could remain in compliance without taking either of those drastic steps; it would, however, be sharing in the cost of installing the extra equipment in the leasing utility. Leased credits facilitate an efficient transition into the new regime of more stringent controls.

Unless the process to determine the level of an effluent or emissions charge includes some automatic means of temporal adjustment, the tendency is for the real rate (adjusted for inflation) to decline over time.[35] This problem is particularly serious in areas with economic growth where increasing real rates would be the desired outcome.

In contrast to emissions trading where ERC prices respond automatically to changing market conditions, emission charges have to be determined by an administrative process. When the function of the charge is to raise revenue for a particular purpose, charge rates will be determined by the costs of

[33] See Feldman and Raufer (1987) and Tietenberg (1989*a*).
[34] Feldman and Raufer (1987).
[35] For further information see Vos (1989) and Sprenger (1989).

achieving that purpose; when the costs of achieving the purpose rise, the level of the charge must rise to secure the additional revenue.[36]

Sometimes that process produces an unintended dynamic. In Japan, for example, the charge is calculated on the basis of the amount of compensation paid to victims of air pollution in the previous year. While the amount of compensation has been increasing, the amount of emissions (the base to which the charge is applied) has been decreasing. As a result unexpectedly high charge rates are necessary in order to raise sufficient revenue for the compensation system.

In countries where the tax revenue feeds into the general budget, increases in the level of the charge require a specific administrative act. Evidently it is difficult to raise these rates in practice, since charges have commonly even failed to keep pace with inflation, much less growth in the number of sources. The unintended result is eventual environmental deterioration.

VI. CONCLUDING COMMENTS

Our experience with economic incentive programmes has demonstrated that they have had, and can continue to have, a positive role in environmental policy in the future. I would submit the issue is no longer *whether* they have a role to play, but rather *what kind* of role they should play. The available experience with operating versions of these programmes allows us to draw some specific conclusions which facilitate defining the boundaries for the optimal use of economic incentive approaches in general and for distinguishing the emissions trading and emission charges approaches in particular.

Emissions trading integrates particularly smoothly into any policy structure which is based either directly (through emission standards) or indirectly (through mandated technology or input limitations) on regulating emissions. In this case emission limitations embedded in the operating licences can serve as the trading benchmark if grandfathering is adopted.

Emissions charges work particularly well when transactions costs associated with bargaining are high. It appears that much of the trading activity in the United States has involved large corporations. Emissions trading is probably not equally applicable to large and small pollution sources. The transaction costs are sufficiently high that only large trades can absorb them without jeopardizing the gains from trade. For this reason charges seem a more appropriate instrument when sources are individually small, but numerous (such as residences or automobiles). Charges also work well as a device for increasing the rate of adoption of new technologies and for raising revenue to subsidize environmentally benign projects.

Emissions trading seems to work especially well for uniformly mixed pollutants. No diffusion modelling is necessary and regulators do not have to worry about trades creating 'hot spots' or localized areas of high pollution concentration. Trades can be on a one-to-one basis.

Because emissions trading allows the issue of who will pay for the control to be separated from who will install the control, it introduces an additional degree of flexibility. This flexibility is particularly important in non-attainment areas since marginal control costs are so high. Sources which would not normally be controlled because they could not afford to implement the controls without going out of business, can be controlled with emissions trading. The revenue derived from the sale of emission reduction credits can be used to finance the controls, effectively preventing bankruptcy.

Because it is quantity based, emissions trading also offers a unique possibility for leasing. Leasing is particularly valuable when the temporal pattern of emissions varies across sources. As discussed above this appears generally to be the case with utilities. When a firm plans to shut down one plant in the near future and to build a new one, leasing credits is a vastly superior alternative to the temporary installation of equipment in the old plant which would be useless when the plant was retired. The useful life of this temporary control equipment would be wastefully short.

[36] While it is theoretically possible (depending on the elasticity of demand for pollution abatement) for a rise in the tax to produce less revenue, this has typically not been the case.

We have also learned that ERC transactions have higher transactions costs than we previously understood. Regulators must validate every trade. When non-uniformly mixed pollutants are involved, the transactions costs associated with estimating the air quality effects are particularly high. Delegating responsibility for trade approval to lower levels of government may in principle speed up the approval process, but unless the bureaucrats in the lower level of government support the programme the gain may be negligible.

Emissions trading places more importance on the operating permits and emissions inventories than other approaches. To the extent those are deficient the potential for trades that protect air quality may be lost. Firms which have actual levels of emissions substantially below allowable emissions find themselves with a trading opportunity which, if exploited, could degrade air quality. The trading benchmark has to be defined carefully.

There can be little doubt that the emissions trading programme in the US has improved upon the command-and-control programme that preceded it. The documented cost savings are large and the flexibility provided has been important. Similarly emissions charges have achieved their own measure of success in Europe. To be sure the programmes are far from perfect, but the flaws should be kept in perspective. In no way should they overshadow the impressive accomplishments. Although economic incentive approaches lose their Utopian lustre upon closer inspection, they have none the less made a lasting contribution to environmental policy.

The role for economic incentive approaches should grow in the future if for no other reason than the fact that the international pollution problems which are currently commanding centre-stage fall within the domains where economic incentive policies have been most successful. Significantly many of the problems of the future, such as reducing tropospheric ozone, preventing stratospheric ozone depletion, moderating global warming, and increasing acid rain control, involve pollutants that can be treated as uniformly mixed, facilitating the use of economic incentives. In addition larger trading areas facilitate greater cost reductions than smaller trading areas. This also augers well for the use of emissions trading as part of the strategy to control many future pollution problems because the natural trading areas are all very large indeed. Acid rain, stratospheric ozone depletion, and greenhouse gases could (indeed should!) involve trading areas that transcend national boundaries. For greenhouse and ozone depletion gases, the trading areas should be global in scope. Finally, it seems clear that the pivotal role of carbon dioxide in global warming may require some fairly drastic changes in energy use, including changes in personal transportation, and ultimately land use patterns. Some form of charges could play an important role in facilitating this transformation.

We live in an age when the call for tighter environmental controls intensifies with each new discovery of yet another injury modern society is inflicting on the planet. But resistance to additional controls is also growing with the recognition that compliance with each new set of controls is more expensive than the last. While economic incentive approaches to environmental control offer no panacea, they frequently do offer a practical way to achieve environmental goals more flexibly and at lower cost than more traditional regulatory approaches. That is a compelling virtue.

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